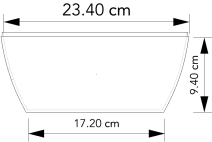
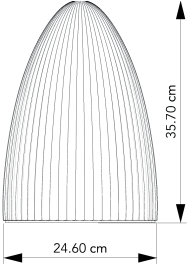
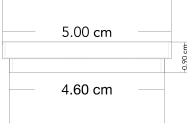
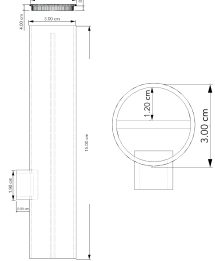
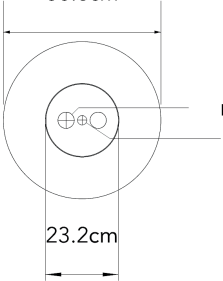
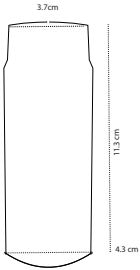
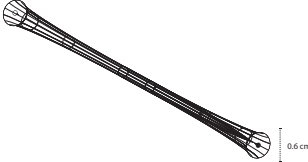
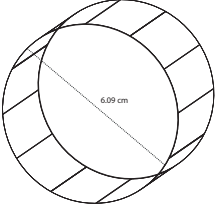


ALGAERITHM ASSEMBLY

PARTS

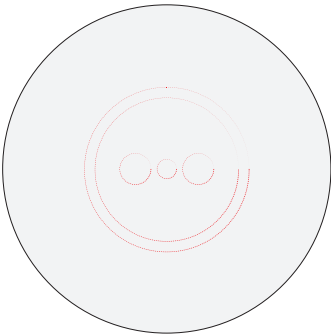
PLA base x 1 	PLA shell x2 	LED stand support x 2 	Column (pump & camera) x1 
3D PRINTED (All STL files available for download on website)			
MDF Plate x1 50.0cm 	Bioreactor x 2 	Air tubes 40" (cut to size) 	Aquarium air pump x1 

Alternative suggestions for materials and notes:

1. All 3D-printed PLA items can be recreated with repurposed material, e.g. shell can be a simple case with a diameter matching the base (50cm).
2. MDF plate can be replaced with lighter-weight repurposed/found material such as cork or marine plywood.
3. Print 2 x PLA shells, one in clear and one in black filament. The black case is to be used while taking a spectrometer reading.

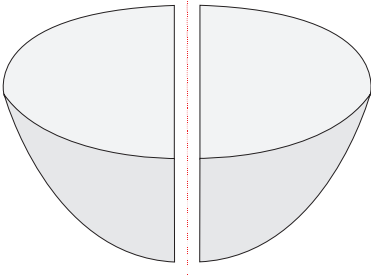
STEPS

1



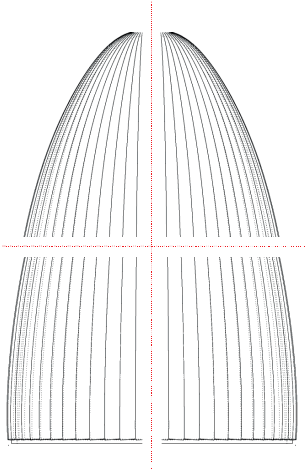
CNC the wooden plate to carve out the contours for the shell, the LED and bioreactor grooves, and the hole for the column.

2



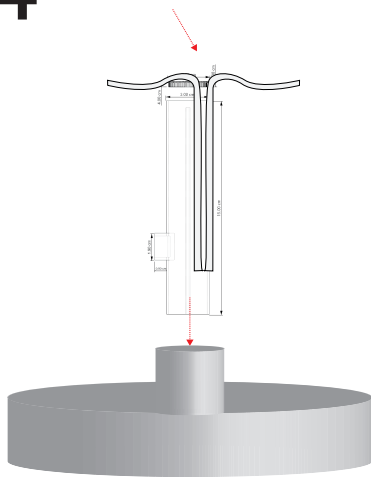
Glue the two parts of the base together.

3



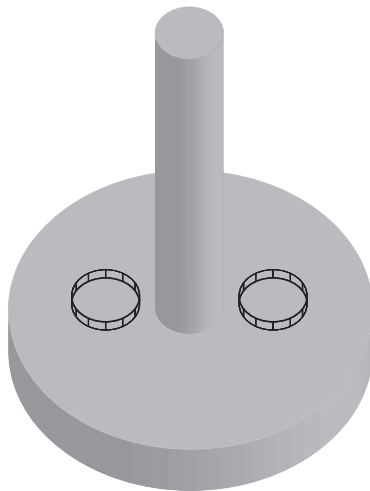
Glue the four parts of the shell together.

4



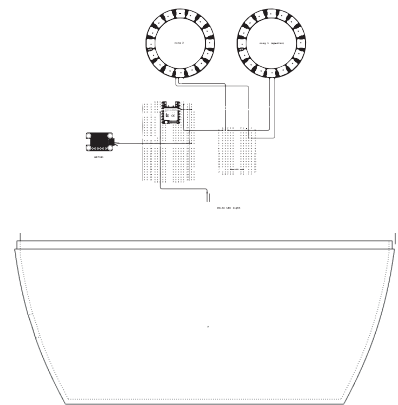
Once printed, install the column and put the tubes through.

5



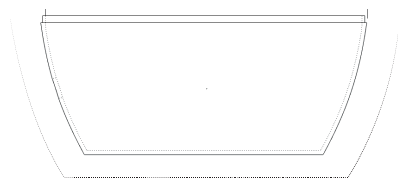
Install the LED supports.

6



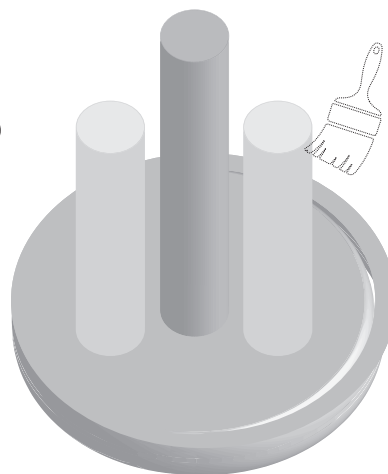
Wire all electronics in the base.
Refer to circuit diagram

7



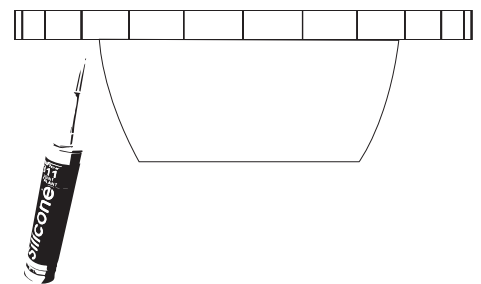
Vacuum-form the base with sheet plastic.

8



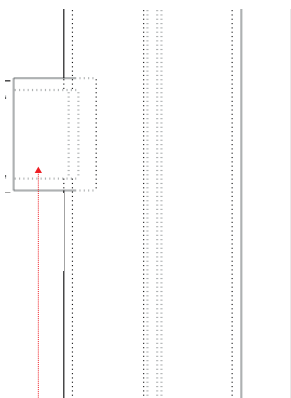
Sand the edges and apply 3 coats of varnish.

9



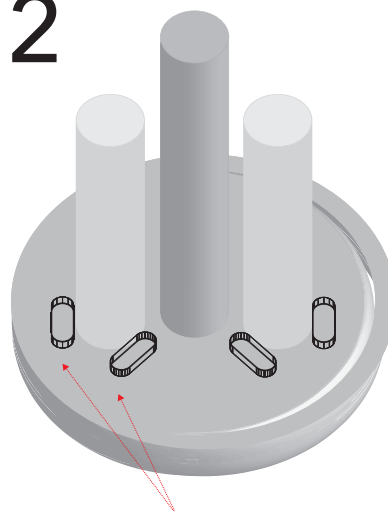
Apply silicone sealant to the base and let rest for 24 hours.

10



Build a support for the spectrometer.
(Can be done with repurposed wood, plastic, or cardboard)

12



If needed, add supports to lock the bioreactors in place.

13



Visit the Algaerithm website for additional support and information on electrical wiring, user app, and more.